

Designing the Multi-level Semantic Web:

① Augmenting the Concept

Accompanying Terms: **a**



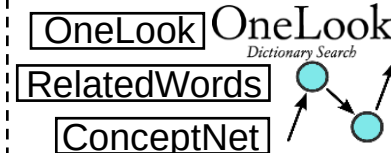
Google books N-gram

Co-occurring Terms: **b**



Google books N-gram

Related Terms: **c**



Similar Terms: **d**

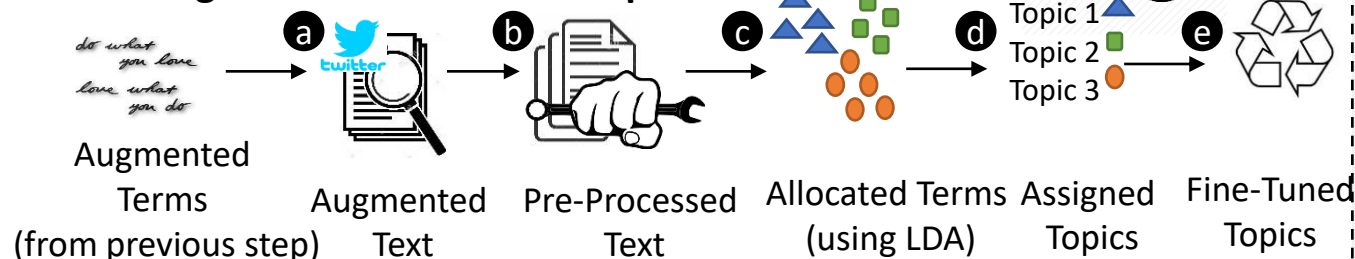
Wikipedia

Google News



Augmented Terms

② Discovering High-Level Topics: Providing a Zoom-out View of Topics



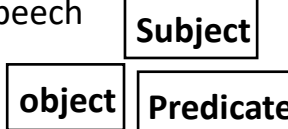
Terms, Topics

③ Extending the Topics: Providing a Zoom-In View of Topics



Augmented Text

Part-Of-Speech Applied



Subject-Object-Predicate Triplets

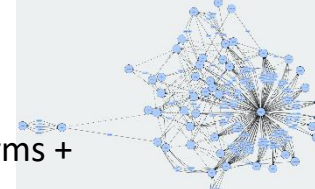
Terms, Topics, Relations

④ Representation



High-Level and Low-Level Terms + Relationships

The Inferred semantic Web



Semantic Web

Augmenting the Dataset:

⑤ Identifying the Gap

USGG



⑥ Augmentation



⑦ Better Performed Object Detector

